

JOONWON LEE

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SUMMARY

Statistics Ph.D. candidate and quantitative researcher specializing in stochastic processes, machine learning, scalable likelihood inference, numerical optimization, and simulation. Builds Python/PyTorch models and statistical diagnostic systems for large, dependent datasets, translating mathematical research into validated computational methods. Experience spans quantitative finance, insurance pricing, and algorithmic market making.

EDUCATION

Ph.D. in Statistics, Rutgers University, Piscataway, NJ Expected Dec 2026 | GPA: 3.8/4.0
M.S. in Statistics, University of Minnesota, Minneapolis, MN Sep 2019 - Sep 2021 | GPA: 3.7/4.0

TECHNICAL SKILLS

Programming and Computing: Python (NumPy, Pandas, SciPy, PyTorch, LightGBM, scikit-learn), SQL/MySQL, R, Linux, Git, AWS EC2, HPC/SLURM

Quantitative Methods: Stochastic processes, time-series and spatiotemporal modeling, MLE/GLS, numerical optimization, Monte Carlo simulation, hypothesis testing, spectral analysis, predictive modeling, GLM/GBM, model validation

EXPERIENCE

JPMorgan Chase - Summer Quantitative Analytics Associate, Quantitative Finance / Risk Modeling Summer 2026

- Developed and validated a Sequential Probability Ratio Test (SPRT) framework for binary risk indicators, translating a business monitoring problem into statistically controlled decision rules.
- Designed simulation-based evaluation of Type I/II error, detection delay, and early-decision tradeoffs, using Brownian-motion approximations to characterize stopping-time uncertainty.

Travelers - Data Science Leadership Program, Business Insurance Pricing Team Summer 2025

- Developed an end-to-end LightGBM pricing pipeline for mid-market property risk, including data preparation, GLM benchmarking, model validation, and performance diagnostics across 2.46M+ records on AWS EC2.
- Presented model diagnostics and feature-attribution findings as pricing and risk-segmentation recommendations for business stakeholders.

QUANTITATIVE RESEARCH AND TRADING PROJECTS

IMC Prosperity Algorithmic Trading Competition - Top 2.39% of Participants Apr 2025

- Designed and evaluated a market-making algorithm with inventory-aware bid/ask quoting, dynamic liquidation rules, and benchmark-driven performance analysis.

Ph.D. Dissertation Research - Scalable Stochastic-Process Inference and Diagnostics Sep 2024 - Present

- Model Development and Calibration:** Developed an advection-aware Vecchia likelihood approximation for nonseparable Gaussian processes, replacing dense covariance operations with fixed-budget conditioning to fit up to 145,008 observations per day.
- Optimization and Model Selection:** Calibrated Matérn and generalized Cauchy dependence models using GLS profiling and L-BFGS optimization and compared competing specifications through empirical diagnostics.
- Simulation and Model Risk:** Built Monte Carlo and scale- and frequency-resolved diagnostics to identify misspecification in smoothness, noise, missingness, and temporal dependence.
- Quantitative Engineering:** Implemented reusable Python/PyTorch likelihood and diagnostic pipelines with dependency-aware data ordering, CPU/GPU execution, cached artifacts, and restartable HPC jobs.

ADDITIONAL

Coursework: Machine Learning, Linear Algebra, Probability Theory, Stochastic Processes, Statistical Computing, Advanced Theory of Statistics I-II, Data Structures and Algorithms, Microeconomics, Econometrics

Languages: Korean (Native); English (Fluent); Chinese (Basic)